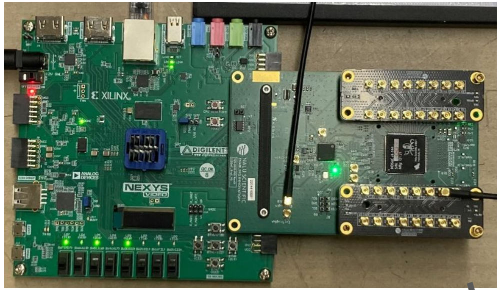
ATAR Testbeam Shared Trigger Prototype DAQ

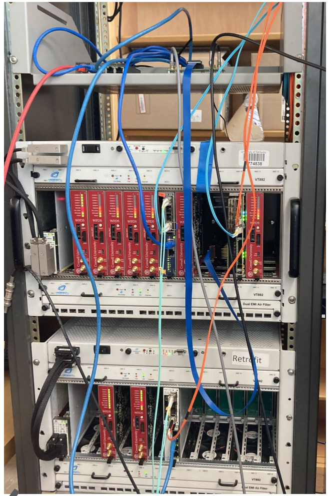
Jack Carlton
University of Kentucky

Current Teststand Setup

- The current hardware we have available in our UKy lab:
  - 1 HDSoCv1 digitizer
  - 2 uTCA crates
  - ~10 working Cornell WFD5s
  - 1 FC7
  - 2 AMC13s
- Provided a shared external trigger to all front end electronics

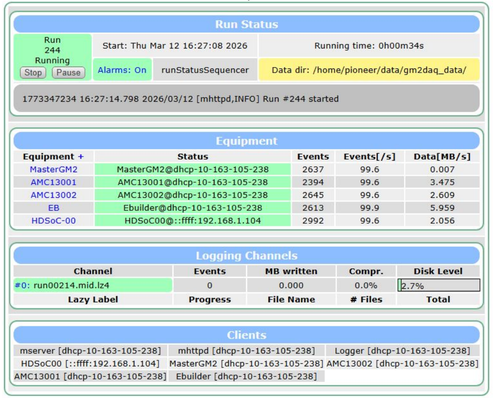


+



HDSoCv1

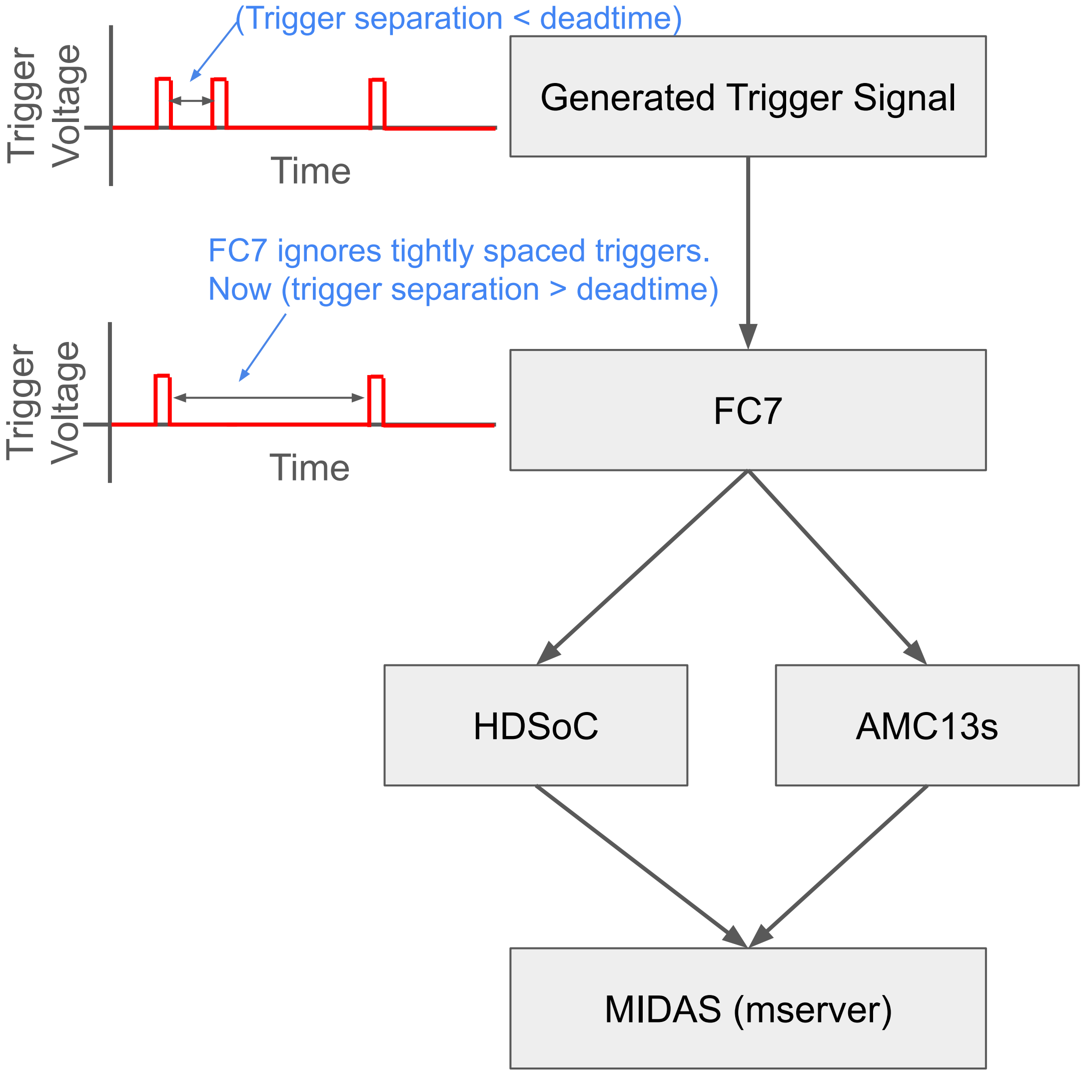
g-2 style teststand



Midas frontends creating data from shared trigger

Shared Trigger Scheme

- Each frontend electronic takes a shared external trigger
  - HDSoCv1 frontend 0
  - AMC13 frontend 0
  - AMC13 frontend 1
- Trigger passes through FC7
  - Ungates signal until run start
  - Gates for programmable “deadtime” during runs
    - <u>Idea:</u> Set this high enough such that all frontends are ready for every event
  - Regates signal after run end
- This scheme will only work with a global external trigger



TODO

- Provide more realistic “random” trigger
  - Using a CAEN function generator to generate periodic pulses with random time seperation
- Incorporate frontend simulators for SAMPic and MuPix
  - Don’t have access to real hardware in our lab, but have simulator frontends available
  - SAMPic should (?) follow a similar shared trigger scheme as the HDSoC
  - Need to better understand MuPix trigger scheme to best design integration
- Put HDSoC on a shared clock
  - Should be possible, need guidance from Nalu/others
- ~~HDSoC v2/v3 integration (?)~~
  - ~~Likely not applicable to ATAR testbeam~~
- Incorporate new systems into DQM framework
  - SAMPic and HDSoC unpackers exist
    - HDSoC has plotting plugins for the webpage already
    - SAMPic does not have plotting plugins written, but this is a short exercise
  - Need to better understand MuPix data output to best design integration

Auxiliary Slides